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RESEARCH ARTICLE

Communication Efficient Cooperative Perception via Codebook-Free Vector Quantization

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ABSTRACT Recent advances in cooperative perception have demonstrated significant performance improvements over single-agent perception. In practice, cooperative perception methods often exchange intermediate multi-channel BEV feature maps at a high frame rate, which can impose a substantial burden on available V2X bandwidth and hinder the practical deployment of such systems. To mitigate bandwidth limitations, we introduce Codebook-Free Quantization for Cooperative Perception (CFQ4CP), a lightweight feature-compression framework. Specifically, a learned neural network, combined with a simple rounding operation, maps continuous latent features to compact integer codes. The discrete representations, instead of high-dimensional feature maps, are then transmitted through V2X communication. Subsequently, these features are reconstructed from the integer codes via a lightweight neural network compression decoder. Unlike existing compression methods that rely on fixed codebooks and nearest-neighbor lookups, our approach generates highly compact integer representations that eliminate the need for codebook alignment between transmitting and receiving agents. We evaluated the proposed framework on two real-world cooperative perception datasets, V2V4Real and DAIR-V2X. Experimental results demonstrate that CFQ4CP achieves perception accuracy competitive with full-precision models, even when each feature vector in the bird's eye view map is quantized to extremely low bitwidths (e.g., 1 or 2 bits), results in reducing the communication costs by thousands of times.

INDEX TERMS Cooperative perception, autonomous driving.

I. INTRODUCTION

By integrating cutting-edge technologies such as artificial intelligence, computer vision, sensor fusion, high-definition mapping, and real-time decision-making algorithms, autonomous vehicles (AVs) are capable of perceiving their environment, planning trajectories, and executing driving maneuvers with minimal or no human intervention. Over the past decade, rapid progress in both hardware and software [1], [2] has advanced autonomous driving toward real-world deployment, prompting extensive academic investigation and industrial investment. Nevertheless, current systems, which are largely based on individual vehicle sensors, often encounter limitations such as blind spots and

incomplete understanding of the dynamic traffic scene. The unreliable perception performance seriously compromises the safety and adoption of autonomous vehicles.

Cooperative perception [3], [4] has emerged as a promising approach to improve environmental understanding by integrating sensory data from multiple autonomous agents to build a more complete and accurate representation of the environment. In contrast to perception systems that rely on a single agent, cooperative perception harnesses the combined sensing capabilities of multiple agents to improve performance and address common challenges such as occlusions, sensor noise, and perceptual ambiguities. This collaborative strategy significantly increases the robustness and reliability of perception systems, enabling their deployment in diverse applications such as autonomous driving and robotics. Recent studies [5], [6], [7], [8], [9], [10] have highlighted the

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substantial benefits of cooperative perception over single-agent approaches, particularly in tasks such as 3D object detection and semantic segmentation.

Cooperative perception relies on vehicle-to-everything (V2X) communication to exchange perception information among connected vehicles and road-side units. The evolution of V2X communication technologies has progressed through several major phases, driven by IEEE standards and successive releases from the 3rd Generation Partnership Project (3GPP). Beginning with IEEE 802.11p and advancing through LTE-V2X to 5G V2X, each generation has delivered notable improvements in data rates and latency [11], [12]. For example, LTE-V2X can support peak data rates of up to 1 Gbps. However, this capacity remains inadequate for many emerging cooperative perception applications. Consider a typical scenario in which a 2D feature map of spatial resolution 256×256 with 256 channels to be transmitted at 10 Hz, the required bandwidth can easily exceed 5 Gbps, far surpassing the capabilities of LTE-V2X systems. Such stringent bandwidth requirements present a critical bottleneck that significantly constrains the practical deployment of cooperative perception. In this paper, we focus specifically on mitigating the communication challenges that arise in cooperative perception for autonomous driving systems.

Existing approaches to reduce communication overhead in cooperative perception can be broadly categorized into two paradigms: (1) feature selection and (2) feature compression. The feature selection paradigm aims to transmit only the most informative or relevant features, such as feature vectors associated with objects of interest. For example, Where2comm [13] and CodeFilling [14] employ confidence-based estimation to sparsify feature maps and transmit only regions deemed important. However, sparse transmission is typically effective only when the sparsity ratio is sufficiently high. When features are quantized to low-bit integers, the overhead associated with sparse encoding becomes non-negligible. We provide a detailed explanation of this limitation in the methodology section. The second paradigm, feature compression, focuses on compressing features prior to transmission and reconstructing them at the receiver. Although less extensively studied, this direction has shown promising results. For example, CodeFilling [14] encodes each feature vector into a compact integer representation, enabling highly efficient transmission while preserving perceptual quality. Despite the progress made in both paradigms, achieving a robust balance between communication efficiency and perception accuracy remains a critical challenge, especially in dynamic and bandwidth-limited environments.

Our method follows the feature compression paradigm. In the field of image/video reconstruction, the Vector Quantized Variational Autoencoder (VQ-VAE) [15] has demonstrated the ability to compress feature maps into discrete integer representations, suggesting its potential as

an effective tool for feature compression. VQ-VAE performs lossy compression by encoding input data into a discrete latent space through vector quantization, where codebooks are employed to facilitate efficient quantization. When applied to cooperative perception, VQ-VAE can preserve high-level semantic information while substantially reducing communication overhead. From this perspective, CodeFilling [14] can be interpreted as a codebook-based Vector Quantized Variational Autoencoder (VQ-VAE) approach for feature compression. It employs a learnable codebook and assigns each encoder output to the nearest embedding via a nearest-neighbor search. Optimizing the codebook aims primarily to reconstruct the input feature maps with high fidelity. However, this fidelity is not strictly necessary for downstream perception tasks. For instance, background regions are also reconstructed during this process, even though they are irrelevant to 3D object detection. Moreover, VQ-VAE models are prone to codebook collapse, a phenomenon in which only a small subset of codebook entries is utilized during training, resulting in reduced codebook capacity and diminished representational expressiveness.

In this paper, we introduce a lightweight feature compression framework to alleviate communication bandwidth constraints in cooperative perception. Unlike conventional vector quantization (VQ)-based approaches that rely on fixed codebooks, our method adopts codebook-free quantization [16], [17], eliminating the memory and computational overhead associated with codebook storage and nearest-neighbor lookups. At the compression encoder, input feature maps are first subjected to channel-wise downsampling and subsequently quantized via a simple rounding operation, transforming each channel into low-bit integers and enabling highly efficient bitstream generation. During training, we employ the Straight-Through Estimator (STE) [18] to backpropagate gradients through the non-differentiable quantization step, ensuring end-to-end optimization. At the compression decoder, the original channel dimension is restored through an upsampling module. Our codebook-free design enables stable end-to-end fine tuning, allowing the system to adapt seamlessly to diverse bandwidth constraints without discrete jumps in performance.

We have conducted extensive experiments on two real-world cooperative perception datasets, V2V4Real [19] and DAIR-V2X [20], to evaluate the effectiveness of the proposed framework. The results demonstrate that our method substantially improves communication efficiency by transmitting compact low-bit integer representations of feature maps while maintaining strong perception performance.

In summary, the contributions of this work include the following:

- We reinterpret the VQ-VAE framework as an effective solution to the communication bandwidth constraints in cooperative perception, thereby enabling a new perspective on efficient feature compression.

- We adopt a codebook-free quantization strategy to replace existing codebook-based compression, eliminating the memory and computational overhead associated with codebook storage and lookup operations.
- Through extensive evaluations, we show that the proposed method achieves performance comparable to its full-precision counterpart by quantizing each vector into only 1 to 2 bits, significantly reducing communication costs.

The remainder of the paper is organized as follows: Section II reviews related work on cooperative perception and communication-efficient representation learning. Section III presents the preliminaries and problem formulation, including Section III-A on single-agent perception backbones and Section III-B on cooperative perception formulation. Section IV introduces the communication-efficient cooperative perception framework, including the overall architecture, the feature selection rationale, codebook-based vector quantization, and codebook-free quantization. Section V provides the experimental evaluation, including dataset descriptions, implementation details, quantitative results, robustness analysis, and qualitative evaluations. Section VI discusses practical considerations and limitations. Finally, Section VII concludes the paper and outlines potential directions for future work.

II. RELATED WORKS

Our work addresses the challenges of bandwidth constraints in cooperative perception. This approach leverages a VQ-VAE architecture to effectively compress the shared feature maps into a cooperative system. This research intersects with several key areas, including cooperative perception, deep feature compression, and VQ-VAE.

A. COOPERATIVE PERCEPTION

Cooperative perception enables multiple agents to share perceptual information through V2X communication, thereby significantly enhancing perception performance compared to single-agent systems. To facilitate algorithm development, both simulated datasets [21], [22] and real-world datasets [19], [20], [23] have been introduced. Considerable progress has been made in addressing key challenges inherent to cooperative perception, including communication bandwidth constraints [6], [13], [14], localization inaccuracies [10], [24], latency and asynchrony [5], [25], [26], [27], and heterogeneity among sensing devices [8], [28], [29]. In addition, recent studies have explored camera-only cooperative perception for 3D object detection [7] and semantic occupancy prediction [30]. Despite these advances, a common limitation of existing approaches is the need to transmit full feature maps, which incurs substantial communication costs. Still, existing communication efficient methods focus on region feature selection [13] or vector quantization based on codebooks [14]. In contrast, this work employs a codebook-free quantization strategy that allows

feature maps to be quantized to extremely low bit widths, making region feature selection unnecessary.

B. DEEP FEATURE COMPRESSION

Deep neural networks have been widely explored for compressing high-dimensional representations into compact codes, enabling efficient storage/transmission while retaining task-relevant information. Early approaches leveraged autoencoders to learn compact latent representations, often incorporating sparsity constraints to enhance compressibility [31], [32]. More recently, end-to-end trainable frameworks have emerged that jointly optimize feature extraction and compression, frequently adopting techniques from learned image compression, such as entropy modeling, quantization-aware training, and context-adaptive coding, to achieve high-rate, low-distortion representations [33], [34], [35], [36]. Nevertheless, the majority of existing methods are designed for general image or video compression and pay comparatively limited attention to applications in cooperative perception.

C. VECTOR QUANTIZED VARIATIONAL AUTOENCODER

Vector-Quantized Variational Autoencoders (VQ-VAE) [15] introduce discrete latent representations by mapping encoder outputs to the nearest entry in a learned codebook. Extensions like VQ-VAE-2 improve generation quality via hierarchical and multi-scale latent structures [37]. Traditional VQ methods are based on an explicit codebook, where each code corresponds to a continuous embedding. Codebooks provide strong per-token expressiveness and integrate naturally with autoregressive priors. However, they can suffer from codebook collapse, uneven code usage, and high memory and lookup costs, especially for large vocabularies. To address these limitations, lookup-free quantization methods have emerged. Examples include finite-scale quantization (FSQ) [17] and factorized or binary discretization approaches [16], [38]. These methods eliminate explicit embedding tables, reducing memory and search costs, and enabling extremely large effective vocabularies. Building on these insights, our approach utilizes codebook-free quantization to achieve efficient cooperative communication perception while maintaining detection performance.

III. PRELIMINARIES AND PROBLEM FORMULATION

This section first reviews the single-agent baseline 3D object detection models used in this paper. Subsequently, we formally define the cooperative perception problem.

A. SINGLE-AGENT PERCEPTION

In this paper, we utilize two LiDAR-based methods, PointPillars [39] and SECOND [40], and one camera-based method, Lift, Splat, Shoot (LSS) [41] as our single-agent backbones.

PointPillars [39] is a highly efficient and widely adopted 3D object detection framework for LiDAR point clouds. Unlike voxel-based methods that quantize space into 3D grids, PointPillars organizes raw point clouds into

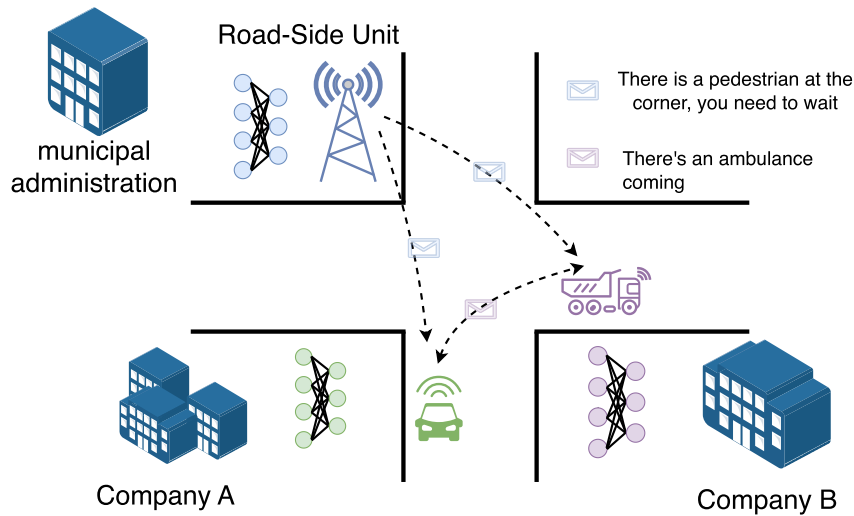


FIGURE 1. Conceptual architecture of cooperative perception. A Road-Side Unit (RSU) functions as a dissemination node, typically deployed at an elevated location to provide a bird's eye view of the roadway. Equipped with sensing and communication capabilities, the RSU and the autonomous vehicles collect and share the perceived information to the other nearby connected vehicles via V2X communication. This cooperative exchange supports real-time decision-making, enhances situational awareness, and enables robust perception and hazard mitigation beyond line-of-sight constraints.

vertical columns, referred to as *pillars*, that span the full height of the scene. Each pillar is processed by a lightweight PointNet-like encoder to extract the per-point features, which are then scattered back into a pseudo-image representation based on the 2D location of the pillar. This pseudo-image is subsequently fed into a standard 2D convolutional backbone (e.g., ResNet) followed by detection heads for bounding box regression and classification. By avoiding expensive 3D convolutions and leveraging mature 2D detection pipelines, PointPillars achieves real-time inference speeds while maintaining competitive detection accuracy, making it suitable for autonomous driving applications.

SECOND [40] is a 3D object detection method that leverages sparse 3D convolutions to efficiently process voxelized LiDAR point clouds. Unlike PointPillars, which projects point features into a dense 2D bird's eye view (BEV) pseudo-image and relies entirely on 2D convolutions, SECOND operates directly in 3D space by partitioning the scene into regular voxels and applying sparse 3D convolutions that activate only on non-empty voxels. This design preserves fine-grained geometric structure and enables richer spatial reasoning compared to the pillar-based height-aggregation in PointPillars.

LSS [41] is a monocular 3D object detection framework that infers 3D scene geometry from 2D image features without explicit depth supervision during training. Unlike LiDAR-based methods such as PointPillars and SECOND, which operate directly on point clouds, LSS leverages only RGB images as input. Its core idea involves three key steps: (1) Lift: per-pixel 2D image features are augmented with a learned depth distribution to produce a set of frustum-based 3D feature volumes; (2) Splat: these 3D features

are projected into a BEV representation via differentiable volumetric pooling; and (3) Shoot: a standard BEV detection head predicts 3D bounding boxes.

All three of these single-agent methods generate an explicit spatial feature representation, where each grid corresponds to a localized feature space in the environment. By exchanging features of the grid through wireless communication, agents can overcome their visual limitations and achieve a more robust perception.

B. COOPERATIVE PERCEPTION FORMULATION

Cooperative perception is a technique that enables multiple agents to collectively perceive a scene by exchanging perceived information through wireless communication, e.g. V2X communication [42]. Fig. 1 presents a conceptual architecture of cooperative perception.

In prior work, cooperative perception methods are commonly categorized into three fusion paradigms: early fusion, intermediate fusion, and late fusion. Early fusion requires the exchange of raw sensor data, which is communication-intensive and insufficiently exploits the computational resources available to individual agents. Late fusion, while model-agnostic, assumes identical perception tasks across agents and typically yields inferior performance [8], [9], [43]. Given these limitations, this paper exclusively focuses on intermediate fusion.

Within the intermediate fusion, two communication strategies are prevalent. Two-round communication involves each agent first acquiring the pose of the ego agent and projecting its raw data into the ego coordinate frame. The intermediate feature maps are then exchanged once they have

been computed. In contrast, one-round communication allows each agent to process its raw data in its own coordinate system and subsequently transmit both its pose and intermediate features simultaneously. A key drawback of the two-round strategy is that it is fully altruistic. Agents cannot obtain perception results in their own coordinate frames. For practical deployment, it is preferable for agents to maintain the ability to operate independently and switch to cooperative mode when needed. Without further mention, this paper considers only the one-round communication setting.

Consider a scenario with N agents in the same scene, each agent $n \in \{1, \dots, N\}$ senses the environment using its sensors and processes the data X_n captured to obtain a feature map F_n . Subsequently, these processed features, along with the agent's pose, are broadcast through a communication network. The feature extraction method remains consistent with the single-agent approach. In the current cooperative perception setting [3], [4], we require each agent to have a spatial feature representation, typically in the form of BEV feature maps. BEV feature maps have clear spatial relevance, which makes it easy to fuse feature maps from distributed autonomous vehicles.

Since the generated single-agent feature maps have a high volume of data, we need to compress the feature maps to fit the limited bandwidth. Here, we use the notation f_{comp} to denote the compression process. The paper focuses on designing the compression process for efficient transmission. The features received from other agents are then projected onto the ego-agent's coordinate system. Finally, the ego agent fuses all features to generate a fused feature map $\hat{F}_{\text{ego}}$ in the ego's coordinate system, via the fusion operator f_{fuse} .

The fused feature map is then fed into a detection head $f_{\text{head}}(\cdot)$ to obtain the final result Y_{ego} . The cooperative perception tasks can be formulated as follows:

$$F_n = f_{\text{backbone}}^n(X_n), \quad n \in \{1, \dots, N\}, \quad (1)$$

$$F_n^q = f_{\text{comp}}(F_n), \quad (2)$$

$$\hat{F}_{\text{ego}} = f_{\text{fuse}}(F_{\text{ego}}, F_{n, n \neq \text{ego}}^q), \quad (3)$$

$$Y_{\text{ego}} = f_{\text{head}}(\hat{F}_{\text{ego}}). \quad (4)$$

Here, $f_{\text{backbone}}^n(\cdot)$ denotes the single-agent feature backbone for the n -th agent. F_n^q denotes the compressed (quantized) representation generated from F_n .

IV. COMMUNICATION EFFICIENT COOPERATIVE PERCEPTION FRAMEWORK

A. OVERALL ARCHITECTURE

As illustrated in Fig. 2, our framework contains 3 parts: the single-agent feature extraction module, a feature compression module, and a feature fusion module. After these modules, the fused feature maps are ready to be sent to the task-specific head.

For each agent n , the single-agent feature backbone f_{backbone}^n processes the raw data and produces BEV features $F_n \in \mathbb{R}^{H_n \times W_n \times C_n}$, where H_n , W_n , and C_n denote

the height, width, and number of channels of the feature map, respectively. To further reduce the transmission bandwidth between agents, we incorporate a feature compression module, which will be described in detail in the next subsection.

In this paper, we employ deformable attention as our fusion mechanism. This choice is motivated by previous work [5], which demonstrates that deformable attention provides a highly effective framework for fusing multi-agent features in cooperative perception. Similar to [44], we first initialize a set of grid-shaped learnable parameters $Q \in \mathbb{R}^{\text{bev_h} \times \text{bev_w} \times \text{bev_c}}$ as BEV queries, where bev_h , bev_w are the spatial dimensions of the BEV plane, each grid $q_{p=(x,y)} \in \mathbb{R}^{1 \times \text{bev_c}}$ within the BEV queries representing the feature within a spatial grid at location $p = (x, y)$. Additionally, positional embeddings are added to BEV queries Q before being sent to the fusion module. Our fusion process in Eq. 3 is based on deformable attention, which can be formulated as:

$$\begin{aligned} & \text{DeformAttn}(q, \{p_n\}_{n=1}^N, \{F_n\}_{n=1}^N) \\ &= \sum_{i=1}^{N_{\text{head}}} \mathcal{W}_i \left[\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{inj} \cdot \mathcal{W}'_i F_n(p_n + \Delta p_{nij}) \right], \quad (5) \end{aligned}$$

where N_{head} is the total number of attention heads, N_{key} is the total number of sampled keys for each head. $\mathcal{W}_i$ and $\mathcal{W}'_i$ are learnable weights. $\mathcal{A}_{inj} \in [0, 1]$ is the predicted attention weight, and is normalized by $\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{inj} = 1$. We use $\Delta p_{nij} \in \mathbb{R}^2$ to denote the predicted offsets relative to the reference point p_n .

B. FEATURE SELECTION STRATEGY

Existing methods [13], [14] typically employ a feature selection strategy aimed at sparsifying feature maps to reduce the communication bandwidth. This strategy relies on an information score generator that produces an information score map from the agent's own BEV feature map, thus quantifying the informativeness of each spatial region based on locally available perception. By transmitting only the most informative spatial features, these approaches can substantially reduce the amount of data exchanged during cooperation. However, the effectiveness of this sparsification strategy is closely related to the overhead required to encode the sparse structure. When each BEV grid is compressed to just a few bits, the overhead of storing and communicating the sparsity mask can become substantial. Given that the sparsity rate is typically greater than 10%, we argue that this overhead can exceed the size of the sparse feature data itself. Motivated by this observation, our approach abandons the conventional feature selection strategy.

C. CODEBOOK-BASED VECTOR QUANTIZATION FOR COOPERATIVE PERCEPTION

Codebook-based Vector Quantization is commonly used to compress shared feature representations. In this setting, both the sender and the receiver share the same codebook,

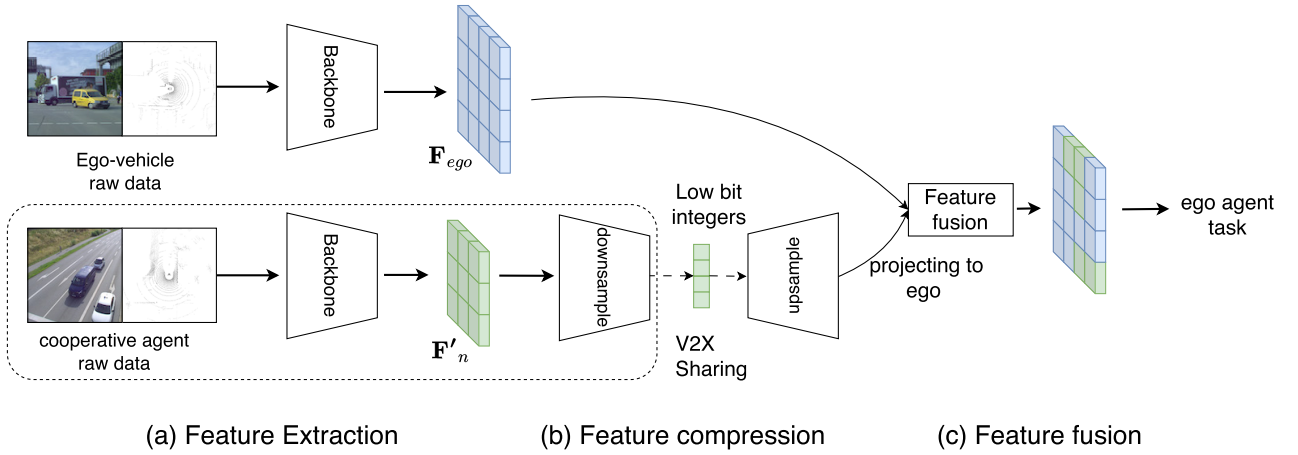


FIGURE 2. Overall architecture. (a) The feature extraction module converts raw sensor data into BEV features in the agent's own coordinate system. (b) To reduce communication costs, the BEV features are first downsampled to d channels and then quantized to L bit values. After the ego vehicle receives the low-bit codes, the features are upsampled back to the original number of channels. (c) Deformable attention is employed to fuse the features effectively.

which consists of a set of learnable prototype vectors designed to approximate the distribution of perceptual features. By transmitting only the indices corresponding to these prototype vectors, the receiver can approximately recover the feature vectors with minimal communication overhead. Formally, this vector is quantized based on its distance to the prototype vectors in the codebook e_k , $k \in 1, \dots, K$ such that each vector $\mathcal{F}_{hw}^n \in \mathbb{R}^{C_n}$ within the BEV feature map $F_n \in \mathbb{R}^{H_n \times W_n \times C_n}$ is replaced by the index of the nearest prototype vector in the codebook, and then the index is transmitted to the receiver. The quantized version of $\mathcal{F}_{hw}^n$ can be expressed as follows,

$$\mathcal{F}_q = e_k \quad \text{where} \quad k = \arg \min_j \|\mathcal{F} - e_j\|. \quad (6)$$

For simplicity, we omit the subscripts n , h , and w in equation 6.

The loss function for codebook-based vector quantization typically consists of two parts: the downstream task loss, such as reconstruction loss in an autoencoder or detection loss in a perception task, and the reconstruction error between the original feature maps and the quantized feature maps. To better align the vector spaces of the codebook and the encoder output, two additional terms are introduced within the reconstruction error component. The first is the codebook loss, which is applied only to the codebook variables and serves to pull the selected codebook vector closer to the encoder's output. The second is commitment loss, which is applied only to the encoder's weights and encourages the encoder output to stay close to the chosen codebook vector, thus preventing it from fluctuating too frequently among different code vectors. The overall training objective is described as:

$$\mathcal{L} = \mathcal{L}_{\text{det}}(\mathcal{F}_q) + \beta \|\mathcal{F}_q - sg(\mathcal{F})\|^2 + \lambda \|\mathcal{F} - sg(\mathcal{F}_q)\|^2, \quad (7)$$

where $\mathcal{L}_{\text{det}}$ denotes the downstream detection loss to be minimized using the quantized feature vectors $\mathcal{F}_q$. β and λ

are hyperparameters that control the reluctance to change the code corresponding to the encoder output. The operator $sg(\cdot)$ refers to a stop-gradient operation that blocks gradients from flowing into its argument, which can be defined as:

$$sg(x) = \begin{cases} x & \text{forward pass} \\ 0 & \text{backward pass} \end{cases} \quad (8)$$

D. CODEBOOK-FREE QUANTIZATION FOR COOPERATIVE PERCEPTION

Existing literatures [16], [17], and [38] shows that codebook-free Vector Quantization has several advantages over codebook-based VQ-VAE. Specifically, codebook-free VQ avoids codebook collapse and under utilization, reduces memory overhead, and offers improved training stability and reconstruction quality compared to codebook-based VQ-VAE.

Let $\mathcal{F} \in \mathbb{R}^C$ denote the input feature vector to be shared. To facilitate efficient representation, we first employ a lightweight compression encoder to project $\mathcal{F}$ into a lower-dimensional space, yielding a feature vector $\mathcal{F}' \in \mathbb{R}^d$, where $d \ll C$. Subsequently, a sigmoid activation function is applied to map the elements of $\mathcal{F}'$ into the interval $(0, 1)$. To quantize the resulting values into L bit precision, we scale the normalized vector by $2^L - 1$ and apply rounding to obtain integer-valued representations within the range $[0, 2^L - 1]$. Finally, a compression decoder maps the quantized vector back to the original C dimensional space. Therefore, f_{comp} is decomposed into a compression encoder, a quantizer, and a compression decoder. The entire pipeline can be formally expressed as follows:

$$\mathcal{F}' = f_{\text{enc}}(\mathcal{F}), \quad (9)$$

$$\tilde{\mathcal{F}} = (2^L - 1) \cdot \sigma(\mathcal{F}'), \quad (10)$$

$$\mathcal{F}'_q = \tilde{\mathcal{F}} + sg(\text{round}(\tilde{\mathcal{F}}) - \tilde{\mathcal{F}}), \quad (11)$$

$$\mathcal{F}_q = f_{\text{dec}}(\mathcal{F}'_q), \quad (12)$$

where $\sigma(\cdot)$ denotes the sigmoid function. Unlike conventional VQ-VAE style compression, our goal is not to reconstruct the original feature vector $\mathcal{F}$ with high fidelity. Instead, CFQ4CP is trained end-to-end with the downstream cooperative perception objective. Compared with equation 7, CFQ4CP uses only one term in the training objective:

$$\mathcal{L} = \mathcal{L}_{\text{det}}(\mathcal{F}_q) \quad (13)$$

and minimizes $\mathcal{L}$ end-to-end. The learned integer representation $\mathcal{F}'_q$ is encouraged to preserve task-relevant information (e.g., object-centric patterns) and is optimized solely through the downstream detection loss, rather than an explicit feature reconstruction loss. During the inference stage of cooperative perception, we only need to transmit the feature vector $\mathcal{F}'_q$. This vector quantization implies an effective codebook size of $2^{L \cdot d}$, resulting in a transmission cost of $L \cdot d$ bits. This representation is a low-bit integer and can be further compressed using lossless compression for V2X communication.

V. EXPERIMENTS

In this section, we first describe the datasets, then outline the implementation details, and finally present the experimental results.

A. DATASETS

1) V2V4REAL [19]

V2V4Real is a real-world vehicle-to-vehicle cooperative perception dataset featuring two vehicles within each scene. The dataset covers a driving area of 410 km, comprising 20K lidar frames. As the RGB data is currently unavailable, our analysis focuses exclusively on LiDAR-based tasks. For the segmentation task, 3D bounding boxes are projected onto the bird's-eye-view plane to generate ground truth targets.

2) DAIR-V2X [20]

DAIR-V2X is another real-world cooperative perception dataset, capturing data from one ego vehicle and one roadside unit within each frame. In total, DAIR-V2X provides a collection of 6,617 frames for cooperative perception, each captured at a rate of 10 Hz with 3D annotations. We adopted the 3d bounding box annotations as in [10], which complemented the missing label to cover 360-degree detection.

B. IMPLEMENTATION DETAILS

For the V2V4Real dataset, the agents with IDs 0 or 1 are randomly selected as the ego agent during the training stage, while the agent with id 0 is always selected as the ego agent during inference. For the DAIR-V2X dataset, the vehicle side is always selected as an ego agent. We use two methods, PointPillars [39] and SECOND [40] for the lidar modality (denoted as L_p and L_s), one method Lift, Splat, Shoot (LSS) [41] for the camera modality (denoted as C). We use the superscript seg to denote the segmentation task.

TABLE 1. Basic agent setting for model-agnostic cooperative perception.

Agent	L_p	L_s	C	L_p^{seg}
Modality	Lidar	Lidar	Camera	Lidar
Downsampling	2x,4x,8x	2x	2x,4x,8x	2x
Method	PointPillars	SECOND	LSS	PointPillars
Task	Detection	Detection	Detection	Segmentation

Since our method supports multi-scale fusion, we evaluate fusion using features at varying scales. For instance, with PointPillars, we utilize features downsampled by factors of 2, 4, and 8, whereas for SECOND, we use a downsampling factor of 2. A summary of the setting of these methods can be found at Tab. 1. We follow the model-agnostic setting as [45], and only finetuning the feature compression module.

For both V2V4Real and DAIR-V2X datasets, we set the detection range as $\pm 100.8m \times \pm 40m$. For the ego agent, we stacked 4 deformable cross-attention layers and 2 deformable self-attention layers as the model-agnostic feature fusion module. Each layer has 8 heads and 8 sampling points. All collaborators in our experiments follow the 1-round communication setting, in which each agent shares the pose and feature map simultaneously. We used the same anchor-based detector head and losses for 3D object detection task as in [8], [10], [13], and [21], with an initial learning rate of 2×10^{-3} to train the detection model. For both DAIR-V2X and V2V4Real datasets, we fine-tuned the models for 20 epochs. We use the weighted cross-entropy with a learning rate of 2×10^{-4} for the segmentation task. The background and foreground weighting values were set at 0.005 and 1, respectively. The 3D object detection task is based on the Average Precision (AP) metric [46] at the specified Intersection-over-Union (IoU) thresholds of 0.5 and 0.7. The segmentation metric is based on the IoU score at the BEV grid level.

C. EVALUATION RESULTS

Tab. 2 shows the cooperative 3D detection performance on the V2V4Real and DAIR-V2X datasets in comparison to benchmarks, including single agent, V2XViT [8], openv2v [21], where2comm [13], and CodeFilling [14]. V2XViT and OpenV2V do not incorporate a communication-aware design and therefore require a large transmission bandwidth. Where2comm is a feature selection method, we report the results using all selected features as a baseline. For CodeFilling, we evaluated multiple codebook sizes using the officially released implementation and observed that larger codebooks generally lead to improved performance, but at the cost of increased communication overhead. For our method, we report the results with only 1 bit representation ($d = 1$ and $L = 1$) for each BEV feature vector, together with fusion results of cooperating with different single-agent methods. Notably, our method demonstrates competitive results, often surpassing existing approaches even when only transmitting 1 bit of data at each bev grid. Compared to full precision

TABLE 2. Comparison of 3D detection performance.

Dataset	V2V4Real		DAIR-V2X		
Metric	AP@0.5	AP@0.7	AP@0.5	AP@0.7	Bitrate (bps) *
Single (L_p)	60.0	43.2	68.6	59.9	-
V2X-ViT	66.2	39.7	70.4	53.1	492M
Openv2v	71.0	45.6	73.3	55.3	861M
Where2comm(max fusion)	76.6	47.7	79.9	65.9	861M
CodeFilling($N_l=2$)*	71.2	44.0	78.4	61.2	246K
CodeFilling($N_l=128$)	72.4	46.7	79.6	63.3	1.72M
$L_p + L_p$	76.0	46.5	79.8	67.0	323K
$L_p + L_s$	76.7	48.7	79.5	66.5	246K
$L_p + L_p^{seg}$	75.0	46.6	79.2	66.5	323K
$L_p + C$	-	-	72.3	60.2	323K

* N_l denotes the codebook size used in CodeFilling. We report results of our method at a compression rate of 1 bit ($d = 1$ and $L=1$), which corresponds to a codebook size of 2.

* Bitrate is measured at 10 Hz, which can be further compressed with lossless compression.

(e.g., floating-point representations with 64 channels), the method reduces the communication costs by thousands of times while maintaining performance. The results show that the transmitted data are highly redundant in the cooperation perception scenario and can be compressed to a very low bit rate.

We further evaluate the impact of different compression rates compared to full-precision features, as shown in Tab. 3. Notably, even with 1-bit compression, our method achieves performance closely approaching that of full-precision representations. With 2-bit compression ($d = 1$ and $L = 2$), the detection accuracy becomes comparable to, or in some cases slightly exceeds, the full-precision fusion results. One anomalous case is the AP@0.7 gain for $L_p + L_p^{seg}$ on V2V4Real under 1-bit compression. We attribute this to configuration sensitivity to localization noise and run-to-run variance, and interpret it as a configuration-specific effect rather than evidence that stronger compression is universally beneficial. These findings suggest that, given an extremely low-bit representation, conventional feature selection may be unnecessary, as the overhead required to encode and transmit a selection mask could outweigh its benefits.

D. ROBUSTNESS AGAINST NOISES

We further investigated the impact of high compression rates on robustness to time delays and pose errors, as shown in Fig. 3 and Fig. 4. The results indicate a similar trend of performance degradation across all models: detection accuracy declines rapidly as noise levels increase. The AP@0.7 metric is more sensitive to these perturbations than AP@0.5. Interestingly, under extreme noise conditions, such as a 500 ms communication delay, the highly compressed model performs comparably, or even slightly better, than the less compressed counterpart. In the evaluated settings, aggressive compression shows robustness trends comparable to those of less compressed models under time-delay and

pose-error perturbations. On the contrary, full-precision models may be more prone to overfitting under ideal, noise-free conditions.

E. QUALITATIVE EVALUATIONS

Fig. 6 and Fig. 5 visualize the quantized feature maps of CodeFilling and our method, respectively. Both approaches successfully highlight the regions of interest. An interesting distinction emerges between the two methods. In CodeFilling, the compressed feature map also preserves background noise, reflecting its objective to maintain the reconstructed feature map as close as possible to the uncompressed representation. In contrast, our method's quantization selectively filters out the background, primarily retaining the objects of interest. This demonstrates that codebook-free quantization focuses on optimizing downstream task performance rather than the fidelity of feature reconstruction. These visualization results align closely with the respective optimization objectives: CodeFilling maintains a dictionary to optimize reconstruction accuracy, whereas our method directly targets the downstream perception task.

Fig. 7 and Fig. 8 visualize the detection results on the V2V4Real dataset and DAIR-V2X dataset, comparing single-agent detection results, CodeFilling [14] and our full precision results. Single-agent detection is limited by a restricted field of view and inherent sensor constraints, resulting in difficulties in handling occlusions. For example, in the V2V4Real dataset, single-agent models frequently fail to detect objects in densely populated urban environments where buildings and other structures block visibility. Both CodeFilling and our method effectively extend the perceptual range beyond the limitations of individual agents. Notably, our approach achieves comparable detection performance to these methods while requiring significantly lower transmission bandwidth.

VI. DISCUSSION AND LIMITATIONS

This section discusses practical considerations, limitations, and potential extensions of CFQ4CP. We analyze how the achieved communication savings translate to end-to-end latency in a realistic V2X pipeline, provide intuition for why extremely low-bit quantization can still retain strong detection performance, discuss the impact of non-ideal V2X channels, and outline the scope of CFQ4CP beyond BEV-based intermediate fusion.

A. FROM COMMUNICATION SAVING TO END-TO-END LATENCY

Our primary objective is to reduce the amount of data exchanged among agents. In a practical V2X system, the delay for the proposed module can be decomposed as $T_{total} = T_{enc} + T_{tx} + T_{dec}$, where T_{enc} and T_{dec} are the feature compression and decompression times, and T_{tx} denotes the wireless transmission delay. For typical cooperative perception pipelines, T_{tx} can dominate when multi-channel BEV feature maps are transmitted at a high

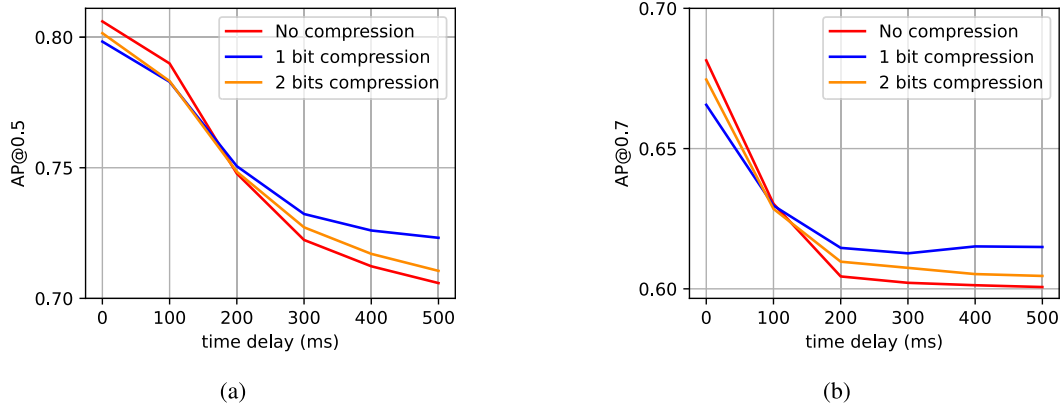


FIGURE 3. Comparison of 3D detection performance on DAIR-V2X datasets under latencies ranging 0-500ms. (a) AP@0.5 (b) AP@0.7.

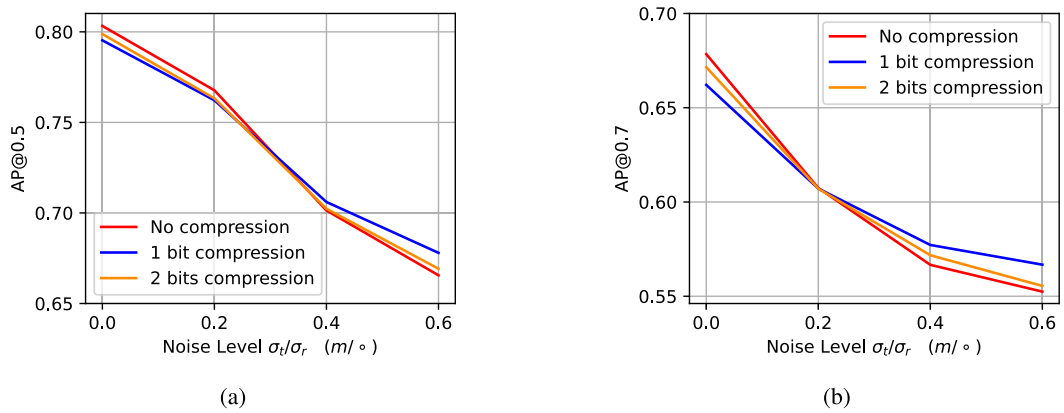


FIGURE 4. Comparison of 3D detection performance on DAIR-V2X datasets under different pose errors (σ_t and σ_r denotes the standard deviation of gaussian noise, the translation and rotation in meters and rotation in degrees separately). (a) AP@0.5 (b) AP@0.7.

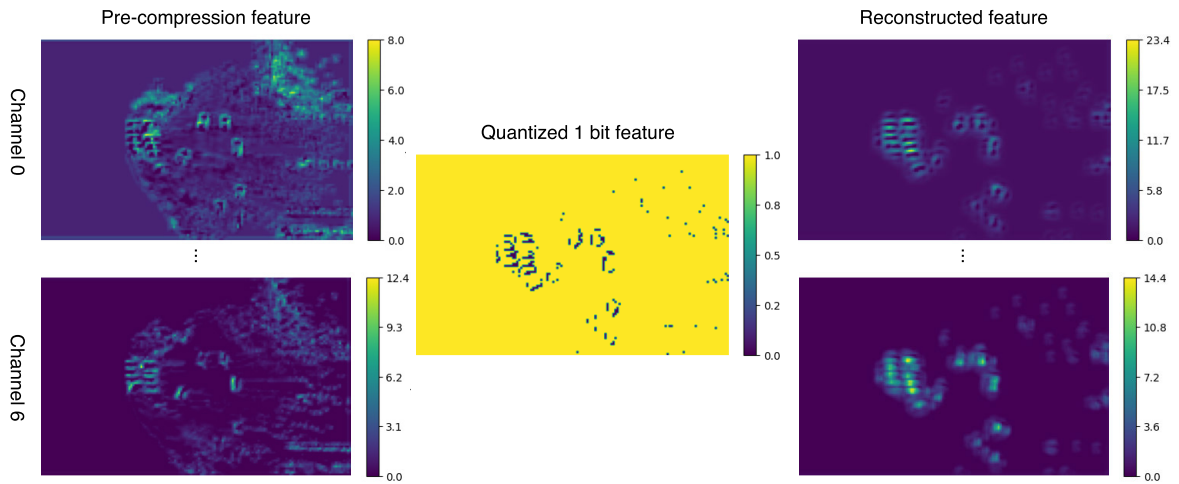


FIGURE 5. Visualization of the feature maps of the $L_p + L_s$ scenario on DAIR-V2X dataset. We show the feature map with a downsampling ratio of 2x. Our method selectively preserves the detection objects while filtering out background information.

frame rate. By quantizing each BEV feature vector to very low bitwidths, CFQ4CP directly reduces the payload size and thus decreases T_{tx} . Meanwhile, the proposed

method consists of lightweight neural networks (typically 3 convolutional layers in both the compression encoder and the compression decoder) and a rounding operation, making

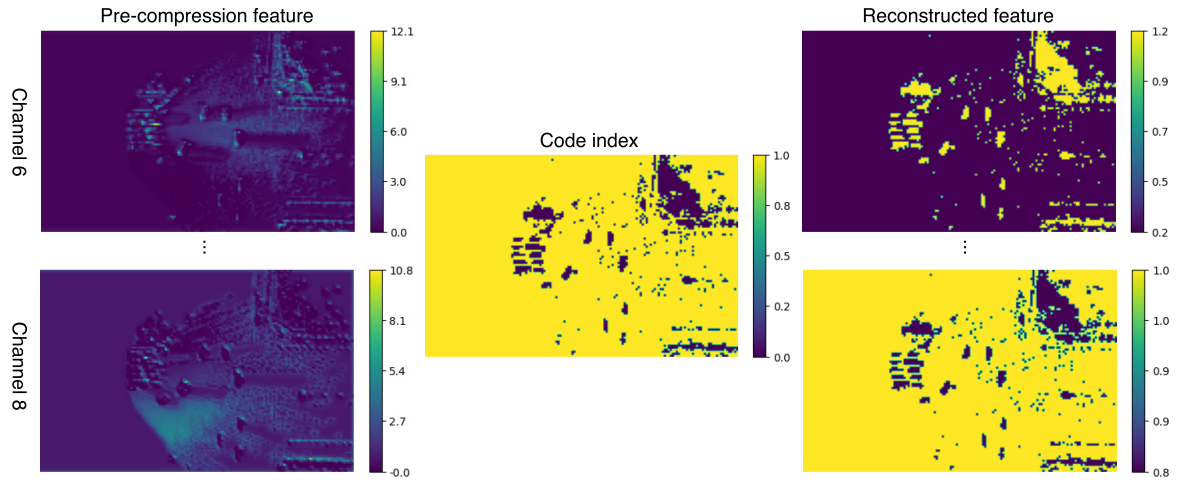


FIGURE 6. Visualization of the feature maps of CodeFilling ($n_l = 2$) on DAIR-V2X dataset. We show the feature map with a downsampling ratio of 2x. The compressed feature maps retain background information.

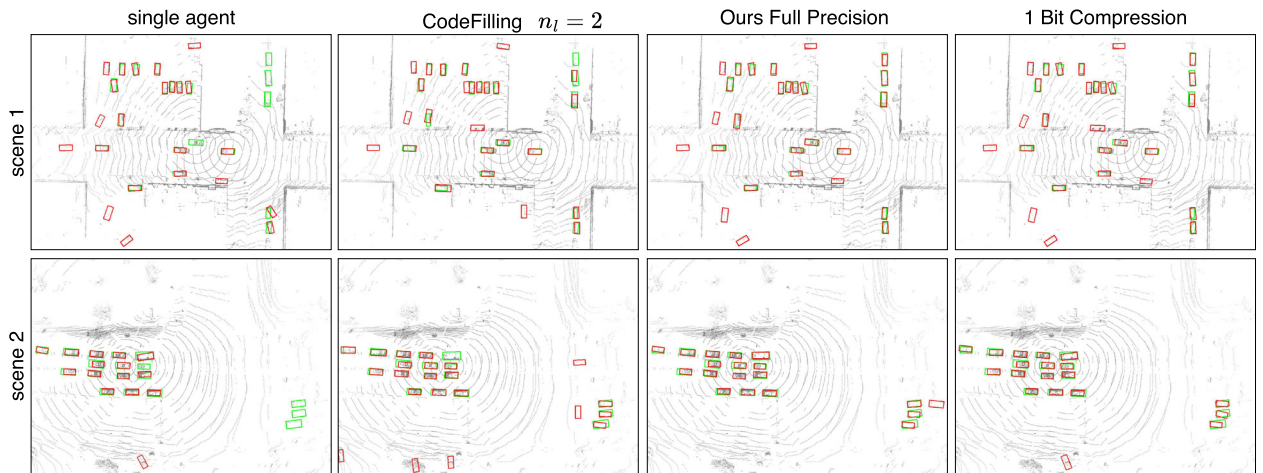


FIGURE 7. Visualization of the detection results on V2V4Real datasets. We show the detection results of single agent, CodeFilling, $L_p + L_p$, and $L_p + L_p^{seg}$. Red boxes: detected, Green boxes: ground-truth. Best viewed zoomed in.

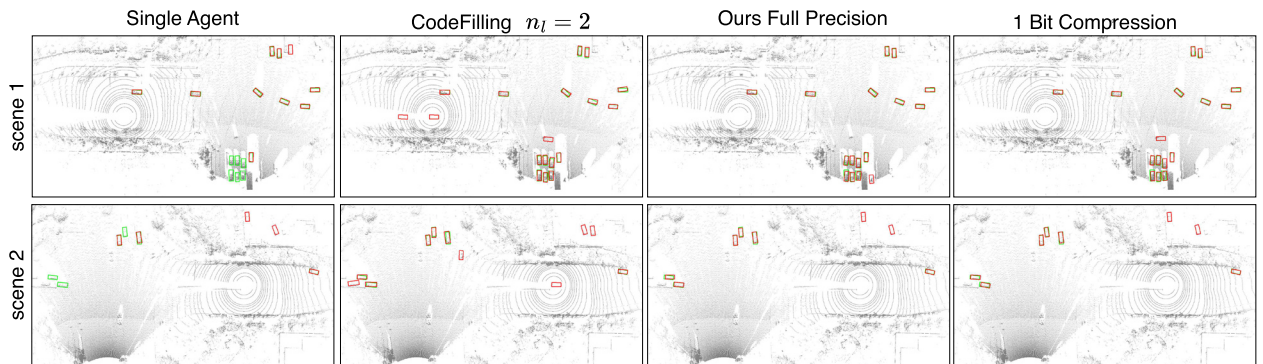


FIGURE 8. Visualization of the detection results on DAIR-V2X datasets. We show the detection results of single agent, CodeFilling, Full Precision result, and 1 bit compression result. Red boxes: detected, Green boxes: ground-truth. Best viewed zoomed in.

T_{enc} and T_{dec} small compared with the transmission savings. Overall, the main latency benefit of CFQ4CP comes

from converting the communication bottleneck into a computation-light local operation.

TABLE 3. Performance cooperating perception with different compression rates.

Rates	No compression		1 bit		2 bits	
Dataset	DAIR-V2X					
Metric	AP@0.5	AP@0.7	AP@0.5	AP@0.7	AP@0.5	AP@0.7
$L_p + L_p$	80.2	67.9	79.8 (↓ 0.4)	67.0 (↓ 0.9)	80.3 (↑ 0.1)	67.7 (↓ 0.2)
$L_p + L_s$	80.1	67.6	79.5 (↓ 0.6)	66.5 (↓ 1.1)	79.9 (↓ 0.2)	67.2 (↓ 0.4)
$L_p + L_p^{seg}$	79.8	67.4	79.2 (↓ 0.6)	66.5 (↓ 0.9)	79.8 (–)	67.0 (↓ 0.4)
$L_p + C$	73.3	59.6	72.3 (↓ 1.0)	60.2 (↑ 0.6)	72.1 (↓ 1.2)	60.7 (↑ 1.1)
Dataset	V2V4Real					
Metric	AP@0.5	AP@0.7	AP@0.5	AP@0.7	AP@0.5	AP@0.7
$L_p + L_p$	77.1	45.8	76.0 (↓ 1.1)	46.5 (↑ 0.7)	77.8 (↑ 0.7)	48.8 (↑ 3.0)
$L_p + L_s$	78.6	47.3	76.7 (↓ 1.9)	48.7 (↑ 1.4)	78.9 (↑ 0.3)	48.9 (↑ 1.6)
$L_p + L_p^{seg}$	75.6	35.6	75.0 (↓ 0.6)	46.6 (↑ 11)	75.4 (↓ 0.2)	41.8 (↑ 6.2)

B. WHY 1–2 BIT QUANTIZATION CAN STILL WORK

Several factors help explain why CFQ4CP can retain strong detection performance even under extreme quantization. First, BEV features produced by perception backbones are redundant and spatially correlated. Neighboring grids and channels are correlated, and many dimensions carry background or weakly-informative content. Second, our quantization module is trained end-to-end with the downstream detection objective, so it is encouraged to preserve task-relevant information rather than faithfully reconstructing all details of the original feature map. Finally, neural networks are able to reconstruct or generate useful representations from heavily compressed information, as shown in Fig. 5, the reconstructed BEV features remain well-aligned with object regions.

C. COMMUNICATION CHANNEL ASSUMPTIONS AND NON-IDEAL V2X

In this work, we assume reliable feature transmission and focus on reducing the payload. However, practical V2X links may still suffer from packet loss and bit errors. Packet loss can lead to missing features from some agents, which may degrade performance, especially in highly occluded scenes. Bit errors can be particularly harmful for low-bit representations, since each flipped bit can cause a relatively larger change. These issues can be mitigated using standard link-layer techniques such as forward error correction (FEC) or by applying training-time augmentation that randomly flips bits to improve robustness. Evaluating CFQ4CP under realistic channel models and designing channel-aware training objectives are important directions for future work.

D. SCOPE BEYOND BEV-BASED INTERMEDIATE FUSION

Although we evaluate CFQ4CP in a BEV-based intermediate fusion pipeline for cooperative 3D object detection,

the proposed codebook-free quantization is not limited to BEV representations. CFQ4CP can be viewed as a task-driven discretization module that can be inserted wherever agents need to transmit intermediate neural features. It is applicable to other feature formats, for example, multi-view image feature maps, voxel/3D grid features, and sparse point-based features. Beyond 3D object detection, CFQ4CP can be extended to other tasks such as BEV semantic segmentation, occupancy prediction, tracking, and motion forecasting, and it can be trained in a multi-task manner with a weighted combination of task losses.

VII. CONCLUSION

This work addresses the critical trade-off between perception accuracy and communication bandwidth in cooperative perception by introducing Codebook-Free Quantization for Cooperative Perception (CFQ4CP). By leveraging a learnable neural network paired with a simple rounding operation, our framework maps continuous latent features to compact integer representations, thereby eliminating the need to transmit high-dimensional feature maps. Extensive experiments on the V2V4Real and DAIR-V2X datasets reveal that cooperative perception features contain significant redundancy. CFQ4CP demonstrates that these features can be quantized to extremely low bitwidths, as low as 1 or 2 bits per BEV feature vector, while achieving detection performance competitive with full-precision models. These results highlight the effectiveness of codebook-free quantization as a viable direction for communication-efficient cooperative perception.

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